

Building a Complete Strategy with Functions

Let's see how these individual functions can be combined into a simple but complete trading strategy:

```
1. def simple_rsi_strategy(prices, rsi_period=14, oversold=30, overbought=70):
2.     """
3.     A simple RSI-based mean reversion strategy.
4.
5.     Args:
6.         prices: List of closing prices
7.         rsi_period: Period for RSI calculation
8.         oversold: Threshold for oversold condition (buy signal)
9.         overbought: Threshold for overbought condition (sell signal)
10.
11.     Returns:
12.         List of positions (1 for long, -1 for short, 0 for flat)
13.     """
14.     positions = [0] # Start with no position
15.
16.     for i in range(rsi_period + 1, len(prices)):
17.         # Get price slice up to current bar (excluding current)
18.         price_slice = prices[:i]
19.
20.         # Calculate RSI using previous prices
21.         rsi_value = calculate_rsi(price_slice, rsi_period)
22.
23.         current_position = positions[-1] # Most recent position
24.
25.         if rsi_value < oversold and current_position <= 0:
26.             # Oversold and not long already - buy
27.             positions.append(1)
28.         elif rsi_value > overbought and current_position >= 0:
29.             # Overbought and not short already - sell
30.             positions.append(-1)
31.         else:
32.             # Maintain current position
33.             positions.append(current_position)
35.     return positions
```